

CASCADE: Self-Reorganizing Knowledge Structures via Truth Pressure and Coherence-Preserving Demotion

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Abstract

Knowledge systems face a fundamental tension: new information that contradicts existing foundations cannot be simply appended (causing incoherence) or substituted (causing information loss). We introduce CASCADE, a self-reorganizing knowledge architecture that resolves this tension through *structured demotion*: when new knowledge with higher *truth pressure* Π contradicts existing foundations, the system *contextualizes* old foundations rather than deleting them, while *promoting* the new knowledge. We define truth pressure as $\Pi = (E \cdot P)/S$ where E is evidence strength, P is explanatory power, and S is Shannon entropy. Knowledge is stratified into three layers (Foundation, Theory, Edge) by Π value, and a four-phase cascade protocol executes reorganization when triggered.

We prove three invariants hold during every cascade event: (1) coherence never decreases, (2) information content never decreases, and (3) total system entropy is non-decreasing. Computational experiments on 1,000 cascade events confirm all three invariants hold at 100% rate. In paradigm shift scenarios, CASCADE eliminates all active contradictions while static systems retain them ($C = 1.000$ vs $C = 0.933$, $p < 10^{-47}$). Ablation shows that removing truth pressure reduces accuracy from 100% to 48%. The complete implementation and all experimental code are open-source.

Keywords: Knowledge Reorganization, Belief Revision, Catastrophic Forgetting, Truth Pressure, Paradigm Shift, Coherence Preservation

1 Introduction

When scientific paradigms shift—Newtonian mechanics giving way to general relativity, classical physics to quantum mechanics—the scientific community does not simply *add* the new theory alongside the old. Nor does it *delete* the old theory entirely. Instead, it *contextualizes*: Newtonian mechanics is demoted from “fundamental law” to “excellent approximation valid at low velocities and weak gravitational fields.” The old knowledge is preserved but reframed, while the new knowledge assumes foundational status.

No existing computational knowledge system replicates this process. Neural networks exhibit catastrophic forgetting when learning contradictory information [McCloskey and Cohen, 1989]. Continual learning methods—elastic weight consolidation [Kirkpatrick et al., 2017], progressive neural networks [Rusu et al., 2016], experience replay [Chaudhry et al., 2019]—mitigate forgetting by either protecting old parameters, adding new capacity, or rehearsing old examples. But all three assume knowledge is *additive and immutable*: new information layers atop old without restructuring. When new knowledge fundamentally contradicts existing foundations, these methods either resist the update (preserving incoherence) or accept it (destroying old knowledge).

The AGM framework for belief revision [Alchourrón et al., 1985] formalizes expansion, contraction, and revision operators but does not provide a mechanism for paradigm-level reorganization where entire hierarchies restructure. Knowledge graphs [Bordes et al., 2013, Speer et al., 2017] organize information but lack principled reorganization triggers.

1.1 Contribution

We introduce CASCADE, a knowledge architecture that formalizes the scientific paradigm shift as a computational operation. Our single core contribution is:

A self-reorganizing knowledge structure with a computable reorganization trigger (Π), a four-phase reorganization protocol, and three provable invariants (coherence, information, entropy preservation).

CASCADE is simple: it has one metric (Π), three layers, one trigger condition, and four phases. We deliberately avoid neural network integration, meta-learning, or consciousness claims. The contribution is the reorganization mechanism itself and the mathematical guarantees that accompany it.

2 Mathematical Framework

2.1 Knowledge Representation

Definition 2.1 (Knowledge Block). A knowledge block B is a tuple (c, E, P, S, L, r) where:

- c : content (a proposition, claim, or representation)
- $E \in [0, 1]$: evidence strength (quality and quantity of empirical support)
- $P \in \mathbb{R}^+$: explanatory power (number and importance of phenomena explained)
- $S \in \mathbb{R}^+$: Shannon entropy (uncertainty in the evidence base)
- $L \in \{F, T, E\}$: layer assignment (Foundation / Theory / Edge)
- r : regime—"universal" or a qualifier like "valid for $v \ll c$ "

Definition 2.2 (Truth Pressure). The truth pressure of block B is:

$$\Pi(B) = \frac{E(B) \cdot P(B)}{\max(S(B), \epsilon)} \quad (1)$$

where $\epsilon = 10^{-6}$ prevents division by zero.

Truth pressure is the ratio of *justified explanatory scope* to *remaining uncertainty*. A block with strong evidence ($E \rightarrow 1$), broad scope (P large), and low uncertainty (S small) has high Π ; speculative claims have low Π .

Proposition 2.3 (Properties of Π). *Truth pressure satisfies:*

1. $\Pi \geq 0$ with equality iff $E = 0$ or $P = 0$.
2. Π is monotonically increasing in E and P , decreasing in S .
3. $\partial\Pi/\partial E = P/S > 0$; $\partial\Pi/\partial S = -EP/S^2 < 0$.
4. For fixed S , Π scales linearly: $\Pi(E + \delta E) = \Pi(E) + (P/S)\delta E$.

Proof. All properties follow directly from the quotient structure of (1). □ □

2.2 Pyramid Architecture

Definition 2.4 (Knowledge Pyramid). A pyramid $\mathcal{P} = (\mathcal{F}, \mathcal{T}, \mathcal{E}, \mathcal{D})$ consists of:

- Foundation layer: $\mathcal{F} = \{B : \Pi(B) \geq \tau_F\}$
- Theory layer: $\mathcal{T} = \{B : \tau_T \leq \Pi(B) < \tau_F\}$

- Edge layer: $\mathcal{E} = \{B : \Pi(B) < \tau_T\}$
 - Dependency graph: $\mathcal{D} \subseteq (\mathcal{F} \cup \mathcal{T} \cup \mathcal{E})^2$
- with default thresholds $\tau_F = 1.5$, $\tau_T = 1.2$.

2.3 Coherence Metric

Definition 2.5 (Active Contradiction). Two blocks B_i, B_j are in *active contradiction* if:

1. B_i and B_j are logically inconsistent ($B_i \wedge B_j \vdash \perp$),
2. both claim universal validity ($r_i = r_j = \text{“universal”}$), and
3. they are in the same domain.

A contradiction is *resolved* if either block has been contextualized (its regime r changed from “universal” to a qualified scope).

Definition 2.6 (Coherence). For knowledge set $\mathcal{K}$ with n blocks:

$$C(\mathcal{K}) = 1 - \frac{|\mathcal{A}(\mathcal{K})|}{\binom{n}{2}} \quad (2)$$

where $\mathcal{A}(\mathcal{K})$ is the set of active contradictions.

3 The Cascade Protocol

3.1 Trigger Condition

A cascade is triggered when new block B_{new} satisfies:

$$\Pi(B_{\text{new}}) > \max_{B \in \mathcal{F} \cap \mathcal{X}} \Pi(B) + \Delta\Pi \quad (3)$$

where $\mathcal{X} = \{B \in \mathcal{F} : B \text{ contradicts } B_{\text{new}}\}$ is the set of conflicting foundations, and $\Delta\Pi \geq 0$ is a hysteresis threshold preventing oscillation.

3.2 Four-Phase Protocol

When (3) is satisfied, CASCADE executes:

Phase 1 — Conflict Detection. Identify $\mathcal{X}$: all foundation blocks that contradict B_{new} .

Phase 2 — Compression. For each $B \in \mathcal{X}$: (a) set $r_B \leftarrow \text{“valid in regime } R\text{”}$ (contextualize); (b) set $S(B) \leftarrow S(B)/\gamma$ where $\gamma \in (0, 1)$ is the compression decay (increasing uncertainty, lowering Π). The block naturally drops to Theory or Edge by its new Π value.

Phase 3 — Expansion. Insert B_{new} into $\mathcal{F}$ with regime “universal.”

Phase 4 — Stabilization. Reassign all blocks to layers based on current Π values. Update dependency graph $\mathcal{D}$.

Algorithm 1: CASCADE Reorganization Protocol

Input: Pyramid $\mathcal{P}$, new block B_{new} , threshold $\Delta\Pi$, decay γ

Output: Updated pyramid $\mathcal{P}'$

```
1  $\mathcal{X} \leftarrow \{B \in \mathcal{F} : B \text{ contradicts } B_{\text{new}} \text{ and } r_B = \text{"universal"}\};$ 
2 if  $\mathcal{X} \neq \emptyset$  and  $\Pi(B_{\text{new}}) > \max_{B \in \mathcal{X}} \Pi(B) + \Delta\Pi$  then
3   foreach  $B \in \mathcal{X}$  do
4      $r_B \leftarrow \text{"limited to regime } R_B\text{"};$                                 // Contextualize
5      $S(B) \leftarrow S(B)/\gamma;$                                           // Increase uncertainty
6    $\mathcal{F} \leftarrow \mathcal{F} \cup \{B_{\text{new}}\};$                                 // Promote new block
7   foreach  $B \in \mathcal{F} \cup \mathcal{T} \cup \mathcal{E}$  do
8      $\text{Reassign } B \text{ to layer matching current } \Pi(B);$ 
9 else
10   $\text{Insert } B_{\text{new}} \text{ at layer matching } \Pi(B_{\text{new}});$ 
11 return  $\mathcal{P}'$ 
```

The key insight: CASCADE *contextualizes* rather than *deletes*. After a cascade, Newtonian mechanics still exists in the pyramid—it has simply been reframed as “valid for $v \ll c$, weak fields” with higher entropy and lower Π .

4 Theoretical Analysis

We prove three invariants that hold during every cascade event.

Theorem 4.1 (Coherence Non-Decrease). *Let $\mathcal{P}$ have coherence $C(\mathcal{P})$. After cascade producing $\mathcal{P}'$:*

$$C(\mathcal{P}') \geq C(\mathcal{P}) \quad (4)$$

Proof. Let $\mathcal{A}$ and $\mathcal{A}'$ denote active contradictions before and after cascade. The cascade adds B_{new} (increasing n by 1) and contextualizes blocks in $\mathcal{X}$ (changing their regime from “universal” to qualified).

Active contradictions involving $\mathcal{X}$ are resolved: for each $B \in \mathcal{X}$, r_B is no longer “universal,” so by Definition 2.5, all contradictions involving B become inactive. Let $\mathcal{R}$ be these resolved contradictions, with $|\mathcal{R}| \geq |\mathcal{X}|$ (each compressed block resolves at least its contradiction with B_{new}).

New active contradictions can only arise from B_{new} contradicting blocks outside $\mathcal{X}$. Since B_{new} ’s contradictions were identified as $\mathcal{X}$ in Phase 1, no new active contradictions are introduced.

Therefore $|\mathcal{A}'| \leq |\mathcal{A}| - |\mathcal{R}| \leq |\mathcal{A}|$. Since $\binom{n+1}{2} > \binom{n}{2}$:

$$C(\mathcal{P}') = 1 - \frac{|\mathcal{A}'|}{\binom{n+1}{2}} \geq 1 - \frac{|\mathcal{A}|}{\binom{n}{2}} = C(\mathcal{P}) \quad (5)$$

□

Theorem 4.2 (Information Preservation). *Let $I(\mathcal{P}) = \sum_{B \in \mathcal{K}} E(B) \cdot P(B)$ be total information content. After cascade:*

$$I(\mathcal{P}') \geq I(\mathcal{P}) \quad (6)$$

Proof. The cascade performs two operations: (1) compression of blocks in $\mathcal{X}$ by increasing $S(B)$, which does not change $E(B)$ or $P(B)$; (2) addition of B_{new} with $E(B_{\text{new}}) > 0$ and $P(B_{\text{new}}) > 0$. Therefore:

$$I(\mathcal{P}') = I(\mathcal{P}) + E(B_{\text{new}}) \cdot P(B_{\text{new}}) > I(\mathcal{P}) \quad (7)$$

□

Theorem 4.3 (Entropy Non-Decrease). *Let $H(\mathcal{P}) = \sum_{B \in \mathcal{K}} S(B)$. After cascade:*

$$H(\mathcal{P}') \geq H(\mathcal{P}) \quad (8)$$

Proof. Compression increases entropy: for $B \in \mathcal{X}$, $S'(B) = S(B)/\gamma > S(B)$ since $\gamma \in (0, 1)$. Addition of B_{new} adds $S(B_{\text{new}}) > 0$. No block’s entropy decreases. Therefore $H(\mathcal{P}') > H(\mathcal{P})$. $\square$ $\square$

Proposition 4.4 (Computational Complexity). *Cascade reorganization has worst-case time complexity $O(|\mathcal{K}|^2)$.*

Proof. Conflict detection: $O(|\mathcal{K}|)$. Compression: $O(|\mathcal{X}|) \leq O(|\mathcal{K}|)$. Layer reassignment: $O(|\mathcal{K}|)$. Coherence verification: $O(|\mathcal{K}|^2)$ pairwise checks. Total: $O(|\mathcal{K}|^2)$, dominated by coherence verification. $\square$ $\square$

Remark 4.5. Cascades are rare events (triggered only on paradigm-level contradictions), so the amortized cost of normal insertions is $O(|\mathcal{K}|)$. In our scaling experiments (Section 5.5), we observe $T \propto N^{2.24}$ empirically, consistent with the theoretical bound.

5 Experimental Validation

All experiments were run with the code in `cascade_real_experiments.py` (available in the repository). Every number below was computed, not estimated. Random seeds are fixed and reported for reproducibility.

5.1 Paradigm Shift Scenario

We simulate a physics paradigm shift: 8 “classical” blocks (moderate evidence, $E \in [0.72, 0.88]$, $P \in [1.3, 1.8]$, $S \in [0.30, 0.55]$) are challenged by 8 “modern” blocks (strong evidence, $E \in [0.90, 0.99]$, $P \in [1.9, 2.8]$, $S \in [0.08, 0.18]$). Each modern block contradicts one classical block in the same domain.

We compare three strategies across $n = 200$ trials with randomized parameters:

Table 1: Paradigm shift results ($n = 200$ trials, 8+8 blocks each).

System	Coherence	Accuracy	Active Contradictions
Static	0.933 ± 0.000	1.000 ± 0.000	8.0 ± 0.0
Additive	1.000 ± 0.000	1.000 ± 0.000	0.0 ± 0.0
CASCADE	1.000 ± 0.000	1.000 ± 0.000	0.0 ± 0.0

Both CASCADE and Additive resolve all contradictions, but through different mechanisms. Additive marks lower- Π blocks as “limited” (a form of override), while CASCADE executes the full four-phase protocol with entropy increase and layer reassignment. The distinction becomes critical in the ablation study (Section 5.4).

5.2 Invariant Verification

We triggered 1,000 cascade events on randomly generated pyramids (5–20 initial blocks, random E , P , S) and measured the three invariants:

Table 2: Invariant verification ($n = 1,000$ cascade events).

Invariant	Preserved	Mean Δ	Min Δ
Coherence $\geq$ (Thm. 4.1)	1000/1000 (100%)	0.000	0.000
Information $\geq$ (Thm. 4.2)	1000/1000 (100%)	$+2.727 \pm 0.361$	$+1.801$
Entropy $\geq$ (Thm. 4.3)	1000/1000 (100%)	$+0.156 \pm 0.035$	$+0.067$

All three invariants hold at 100% rate across all tested cascade events, confirming Theorems 4.1–4.3.

5.3 Sequential Learning

We test 30-step knowledge accumulation sequences with 35% contradiction rate across 5 domains ($n = 200$ trials):

Table 3: Final coherence after 30-step sequential learning ($n = 200$).

System	Final Coherence	vs. CASCADE
Static	0.980 ± 0.006	$t = 19.12, p = 5.9 \times 10^{-47}, d = 0.95$
Additive	0.992 ± 0.004	$t = -21.14, p = 9.0 \times 10^{-53}, d = -1.28$
CASCADE	0.986 ± 0.006	—

CASCADE significantly outperforms Static ($p < 10^{-46}$, $d = 0.95$). The Additive system achieves slightly higher coherence in this experiment because it aggressively contextualizes *any* lower-II contradicting block, even when the challenger does not meet the cascade threshold. This trades information fidelity for coherence—a design choice CASCADE intentionally avoids. CASCADE only reorganizes when the evidence genuinely warrants it (when (3) is satisfied), preserving old knowledge at its natural layer when the challenge is insufficient.

5.4 Ablation Study

We remove components one at a time from CASCADE on the paradigm shift scenario ($n = 200$):

Table 4: Ablation study ($n = 200$ trials).

Configuration	Coherence	Accuracy	Contradictions
Full CASCADE	1.000 ± 0.000	1.000 ± 0.000	0.0
No cascade (static)	0.933 ± 0.000	1.000 ± 0.000	8.0
No truth pressure	0.945 ± 0.010	0.481 ± 0.178	6.7
No contextualization	0.933 ± 0.000	1.000 ± 0.000	8.0
Binary layers (F/E only)	1.000 ± 0.000	1.000 ± 0.000	0.0

The critical finding: **removing truth pressure drops accuracy from 100% to 48.1%** (essentially random). Without II, the system cannot distinguish which block should become foundational and which should be contextualized. Removing contextualization entirely produces the same result as Static—8 active contradictions retained.

5.5 Computational Scaling

Table 5: Scaling behavior (mean $\pm$ std over 3 runs).

N	Time (s)	Cascade Events
10	0.0001	0
50	0.0004	1
100	0.002	5
500	0.121	20
1,000	0.914	40
2,500	12.990	—

Empirical scaling: $T \propto N^{2.24}$ ($R^2 = 0.967$), consistent with the theoretical $O(N^2)$ bound from Proposition 4.4. The slight super-quadratic behavior is due to Python interpreter overhead; we expect $O(N^2)$ in optimized implementations.

6 Falsifiable Predictions

Every major claim in this paper can be disconfirmed:

Table 6: Falsifiable predictions with explicit disconfirming criteria.

Claim	Prediction	Falsified If
Coherence invariant	$C(\mathcal{P}') \geq C(\mathcal{P})$ for all cascades	Any cascade where C decreases
Information invariant	$I(\mathcal{P}') \geq I(\mathcal{P})$ for all cascades	Any cascade where I decreases
Truth pressure necessity	Ablation without Π : accuracy $< 60\%$	Accuracy $\geq 80\%$ without Π
Scaling	$T \propto O(N^k)$, $k \leq 3$	$k > 3$ in practice
Threshold robustness	Coherence stable for $\Delta\Pi \in [0, 1]$	Coherence varies $> 10\%$ across range

7 Limitations

We are explicit about what CASCADE does not do and where it may fail.

Quadratic scaling. Coherence verification is $O(N^2)$. For systems with $> 10^4$ blocks, approximate methods (e.g., sampling-based coherence estimation) will be needed.

Contradiction detection is assumed. We assume a function $\text{contradicts}(B_i, B_j) \rightarrow \{0, 1\}$ exists. In practice, detecting logical contradiction between natural language claims is an open NLP problem. CASCADE provides the reorganization mechanism *given* contradiction information.

Threshold sensitivity. While coherence is stable across $\Delta\Pi \in [0, 2]$, the cascade *frequency* depends on threshold choice. Too low: frequent unnecessary reorganization. Too high: failure to reorganize when warranted. Domain-specific tuning is needed.

No neural integration. CASCADE operates on symbolic knowledge blocks, not neural network weights. Integrating CASCADE with gradient-based learning (e.g., using Π to modulate EWC Fisher information) is future work.

Single-paradigm experiments. Our experiments use synthetic knowledge scenarios. Evaluation on real-world knowledge bases (e.g., scientific literature, medical guidelines) with ground-truth paradigm shifts is needed.

Layer count. We use three layers with fixed thresholds. Whether continuous Π -based stratification outperforms discrete layers is untested.

8 Related Work

Continual learning. EWC [Kirkpatrick et al., 2017] and SI [Zenke et al., 2017] add quadratic penalties to protect important parameters. Progressive networks [Rusu et al., 2016] add capacity. PackNet [Mallya and Lazebnik, 2018] prunes and packs. Replay methods [Chaudhry et al., 2019, Shin et al., 2017] rehearse old data. All assume knowledge is additive; CASCADE reorganizes structure.

Belief revision. The AGM framework [Alchourrón et al., 1985] defines expansion (+), contraction (−), and revision (*) operators. CASCADE’s compression is a form of contraction that preserves the contracted belief in a qualified form. The trigger condition (3) extends AGM with a quantitative criterion for when revision should occur.

Knowledge graphs. TransE [Bordes et al., 2013], R-GCN [Schlichtkrull et al., 2018], and ConceptNet [Speer et al., 2017] organize relational knowledge but lack reorganization triggers. Temporal KGs [Trivedi et al., 2017] handle updates but not structural paradigm shifts.

Truth maintenance. JTMS and ATMS [Doyle, 1979, de Kleer, 1986] track justifications for beliefs but do not reorganize hierarchies based on evidence quality.

9 Conclusion

We presented CASCADE, a self-reorganizing knowledge architecture that formalizes how scientific paradigm shifts work: through structured demotion (contextualization) of old foundations and evidence-based promotion of new ones. The mechanism is simple—one metric, three layers, four phases—with three provable invariants confirmed at 100% rate across 1,000 cascade events. The ablation study demonstrates that truth pressure is not optional: without it, reorganization accuracy drops to chance level.

CASCADE is not a complete solution to knowledge management. It requires external contradiction detection, operates on symbolic blocks rather than neural weights, and has quadratic scaling. But it provides a mathematically grounded mechanism for the specific problem of *knowledge reorganization under paradigm shift*—a problem that, to our knowledge, no existing system addresses with formal guarantees.

All code is open-source at <https://github.com/Lycheetah/cascade-framework>.

Reproducibility

Every experimental number in this paper comes from `cascade_real_experiments.py`, runnable with `python cascade_real_experiments.py` using only NumPy and SciPy. Seeds are fixed. We encourage adversarial testing: if you can find a cascade event that violates any invariant, the theory is falsified.

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